project (**π**), select (**σ**), join (⋈)

Suppose we have the following 4 relations:

HOTEL (<u>hotel_no</u>, hotel_name, hotel_city)
ROOM (<u>room_no</u>, <u>hotel_no</u>, room_type, room_price)
BOOKING (<u>hotel_no</u>, <u>guest_no</u>, <u>bdate_from</u>, bdate_to, <u>room_no</u>)
GUEST (<u>guest_no</u>, guest_name, guest_address)

Write the relational algebra for the following queries (your answer **must** show an understanding of query efficiency i.e. you must not make use of unnecessary joins, nor carry attributes and tuples up through the query which are not necessary):

1. List the number and name for all hotels
   $\pi_{\text{hotel_no, hotel_name}}(\text{HOTEL})$

2. List all single rooms with a price below $50
   $\sigma_{\text{room_type="single"} \wedge \text{room_price<50}}(\text{ROOM})$

3. List the numbers and names of all hotels in Melbourne
   $\sigma_{\text{hotel_city="Melbourne"}}(\pi_{\text{(hotel_no, hotel_name, hotel_city}}(\text{HOTEL}))$

4. List all numbers and names of hotels which have a presidential suite room
   $\pi_{\text{hotel_name, hotel_no}}(\sigma_{\text{room_type="presidential suite"}}(\text{ROOM}) \bowtie \text{HOTEL})$

5. List the price and type of all rooms at the Grosvenor Hotel
   $\pi_{\text{room_type, room_price}}(\sigma_{\text{hotel_name = "Grosvenor Hotel"}}(\text{ROOM}) \bowtie \text{HOTEL})$

6. List all numbers, names, and addresses of guests currently staying in deluxe room of any hotel (assume that if the guest has a tuple in the BOOKING relation, then they are currently staying in the hotel)
   $\pi_{\text{guest_no, guest_name, guest_address}}(\text{GUEST} \bowtie (\sigma_{\text{hotel_type="deluxe"}}(\text{ROOM} \bowtie \text{BOOKING})))$

7. List all numbers, names, and addresses of guests currently staying at the Grosvenor Hotel (assume that if the guest has a tuple in the BOOKING relation, then they are currently staying in the hotel)
   $\pi_{\text{guest_no, guest_name, guest_address}}(\text{GUEST} \bowtie (\text{BOOKING} \bowtie (\sigma_{\text{hotel_name = "Grosvenor"}}(\text{HOTEL}))))$

project (**π**), select (**σ**), join (⋈)

Considers these four relations:

CUSTOMER (<u>cust_id</u>, cust_name, cust_address)
PRODUCT (<u>prod_id</u>, prod_desc, prod_unitprice, prod_stock)
STAFF (<u>staff_name</u>, staff_position)
SALE (<u>cust_id</u>, <u>sale_date</u>, <u>prod_id</u>, sale_qty, sold_by)

*Note that sold by value is the *name of staff who made the sale*
also refer to the following tables as seen in a RDBMS

CUSTOMER

cust_id	cust_name	cust_address
111	Clive	India Rd
112	Clark	Kent St
113	Charles	Windsor Av
114	Cilla	Black St

PRODUCT

prod_id	prod_desc	prod_unitprice	prod_stock
K3	Knife Set	$17.95	105
K5	Ladle	$6.95	0
K11	Scraper	$0.95	66
L12	Rack	$22.95	0
L3	Table	$399.50	4
L6	Stool	$17.95	13

STAFF

staff_name	staff_position
Clark	Manager
Simon	Clerk
Steve	Packer
Sean	Clerk
Sorcha	Director
Charles	Clerk

SALE

cust_id	sale_date	prod_id	sale_qty	sold_by
112	20170311	K3	6	Simon
114	20170121	K11	1	Simon
114	20170123	K11	1	Simon
113	20161130	L12	5	Sorcha
114	20170228	L12	1	Sean
113	20161129	K3	2	Sean

Using Relational Algebra, answer the following queries. You must represent your answer in symbolic notation and where a query has several solutions, your answer must represent the most efficient solution.

1. List ids, names of customers and descriptions of products bought by the customer. How many tuples will be returned by the relational algebra query that you have constructed as your answer?

$\boldsymbol{\pi}_{\text{cust_id, cust_name, prod_desc}}(\text{CUSTOMER} \bowtie \text{SALE} \bowtie \text{PRODUCT})$

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2. List all names which are shared by customers and staff

$\boldsymbol{\pi}_{\text{cust_name}}(\text{CUSTOMER}) \cap \boldsymbol{\pi}_{\text{staff_name}}(\text{STAFF})$

3. List ids and descriptions of products that haven't been sold

$\boldsymbol{\pi}_{\text{prod_id, prod_desc}}(\text{PRODUCT}) - \boldsymbol{\pi}_{\text{prod_id, prod_desc}}(\text{SALE} \bowtie \text{PRODUCT})$

4. List names of clerks who don't have any sales yet

$\boldsymbol{\pi}_{\text{staff_name}}(\boldsymbol{\sigma}_{\text{staff_position="Clerk"}}(\text{STAFF})) - \boldsymbol{\sigma}_{\text{sold_by}}(\text{SALE})$

5. List categories (positions) of staff who have made sales

$\boldsymbol{\pi}_{\text{staff_ position}}(\text{STAFF} \bowtie (\boldsymbol{\pi}_{\text{sold_by}}(\text{SALE})))$